10 Microservice APIs in Kubernetes

This chapter covers

- Deploying an API to Kubernetes
- Hardening Docker container images
- Setting up a service mesh for mutual TLS
- Locking down the network using network policies
- Supporting external clients with an ingress controller

In the chapters so far, you have learned how to secure user-facing APIs from a variety of threats using security controls such as authentication, authorization, and rate-limiting. It's increasingly common for applications to themselves be structured as a set of *microservices*, communicating with each other using internal APIs intended to be used by other microservices rather than directly by users. The example in figure 10.1 shows a set of microservices implementing a fictional web store. A single user-facing API provides an interface for a web application, and in turn calls several backend microservices to handle stock checks, processing payment card details, and arranging for products to be shipped once an order is placed.

> DEFINITION A ***microservice*** is an independently deployed service that is a component of a larger application. Microservices are often contrasted with